

# Doxorubicin loading into milk and mesenchymal stem cells' extracellular vesicles as drug delivery vehicles

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## Abstract

Extracellular vesicles (EVs) have great potential as drug delivery vehicles. While mesenchymal/stromal stem cell (MSC) conditioned medium (CM) and milk are potentially safe and scalable sources of EVs for this purpose, the suitability of MSC EVs and milk EVs as drug delivery vehicles has never been compared and so was the objective of this study. Here EVs were separated from MSCs' CM and from milk and were characterised by nanoparticle tracking analysis, transmission electron microscopy, total protein quantification, and immunoblotting. An anti-cancer chemotherapeutic drug, doxorubicin (Dox), was then loaded into the EVs by one of three methods: by passive loading or by active loading by either electroporation or sonication. Dox-loaded EVs were analysed by fluorescence spectrophotometer, high-performance liquid chromatography (HPLC), and imaging flow cytometer (IFCM). Our study showed that EVs were successfully separated from the milk and MSC CM, with significantly ( $p < 0.001$ ) higher yields of milk EVs/mL starting material compared to MSC EVs/mL of starting material. Using a fixed amount of EVs for each comparison, electroporation achieved significantly more Dox loading when compared to passive loading ( $p < 0.01$ ). Indeed, of 250  $\mu\text{g}$  of Dox made available for loading, electroporation resulted in  $90.1 \pm 12 \mu\text{g}$  of Dox loading into MSC EVs and  $68.0 \pm 10 \mu\text{g}$  of Dox loading into milk EVs, as analysed by HPLC. Interestingly, compared to the passive loading and electroporation approach, after sonication significantly fewer CD9<sup>+</sup> EVs/mL ( $p < 0.001$ ) and CD63<sup>+</sup> EVs/mL ( $p < 0.001$ ) existed, as determined by IFCM. This observation indicates that sonication, in particular, may have detrimental effects on EVs. In conclusion, EVs can be successfully separated from both MSC CM and milk, with milk being a particularly rich source. Of the three methods tested, electroporation appears to be superior for achieving maximum drug loading while not causing damage to EV surface proteins.